PMHD Ooko Draft

Gruop 16

2026-04-14

```
# -----------------------------
# 2. Read the data
# -----------------------------
outcome <- read.csv("C:/Users/guest441/Desktop/Hasselt/PMHD/G16.outcome.data.csv", stringsAsF
actors = FALSE)
size <- read.csv("C:/Users/guest441/Desktop/Hasselt/PMHD/G16.size.data.csv", stringsAsFactors
= FALSE)
soil <- read.csv("C:/Users/guest441/Desktop/Hasselt/PMHD/G16.soil.data.csv", stringsAsFactors
= FALSE)
```

```
# -----------------------------
# 3. Clean and prepare variables
# -----------------------------
outcome <- outcome %>%
  mutate(
    climate  = factor(climate),
    species  = factor(species),
    location = factor(location),
    ecotr_id = factor(ecotr_id),
    tree_id  = factor(tree_id),
    pear_id  = factor(pear_id)
  )

# Set reference levels
# Climate reference = Scenario 1 (control/reference condition)
# Species reference = Conference (change if you prefer the other one)


outcome$climate <- relevel(outcome$climate, ref = "Scenario 1")
outcome$species <- relevel(outcome$species, ref = "Conference")

unique(outcome$climate)
```

```
## [1] Scenario 1 Scenario 2 Scenario 3 Scenario 4
## Levels: Scenario 1 Scenario 2 Scenario 3 Scenario 4
```

```
unique(outcome$species)
```

```
## [1] Conference Doyenne
## Levels: Conference Doyenne
```

```r
# ----------------------------
# 4. Basic exploratory checks
# ----------------------------
options(na.print = "NA")

# Sample sizes by climate and species
sample_table <- table(outcome$climate, outcome$species)
sample_table
```

```
##
##              Conference Doyenne
##   Scenario 1        189     160
##   Scenario 2        311     262
##   Scenario 3        220     201
##   Scenario 4        191     161
```

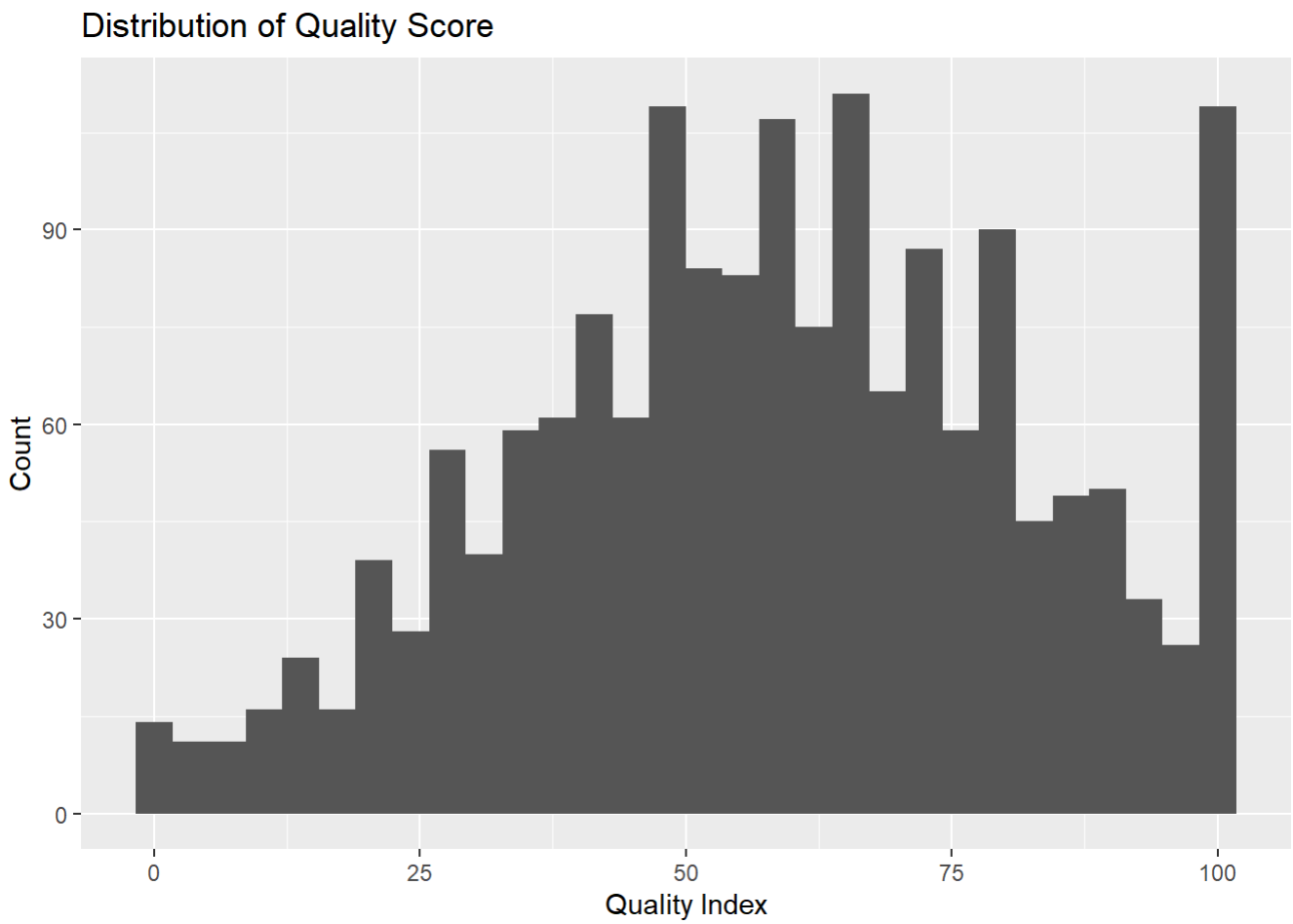
```r
# Number of pears per tree
pear_counts <- outcome %>%
  count(ecotr_id, tree_id, climate, species, name = "n_pears") %>%
  arrange(desc(n_pears))

head(pear_counts, 20)
```

```
##    ecotr_id tree_id    climate    species n_pears
## 1         2       7 Scenario 2 Conference      45
## 2         2       9 Scenario 2 Conference      41
## 3         6      31 Scenario 2 Conference      40
## 4         2      11 Scenario 2 Conference      37
## 5         6      33 Scenario 2 Conference      37
## 6         2      10 Scenario 2    Doyenne      35
## 7         6      36 Scenario 2    Doyenne      34
## 8        10      57 Scenario 2 Conference      34
## 9         3      15 Scenario 3 Conference      32
## 10       10      58 Scenario 2    Doyenne      32
## 11       10      59 Scenario 2 Conference      31
## 12        7      37 Scenario 3 Conference      30
## 13        6      32 Scenario 2    Doyenne      29
## 14       10      56 Scenario 2    Doyenne      29
## 15       10      60 Scenario 2    Doyenne      29
## 16        3      13 Scenario 3 Conference      28
## 17        7      42 Scenario 3    Doyenne      28
## 18        6      34 Scenario 2    Doyenne      27
## 19        9      53 Scenario 1 Conference      27
## 20        4      21 Scenario 4 Conference      26
```

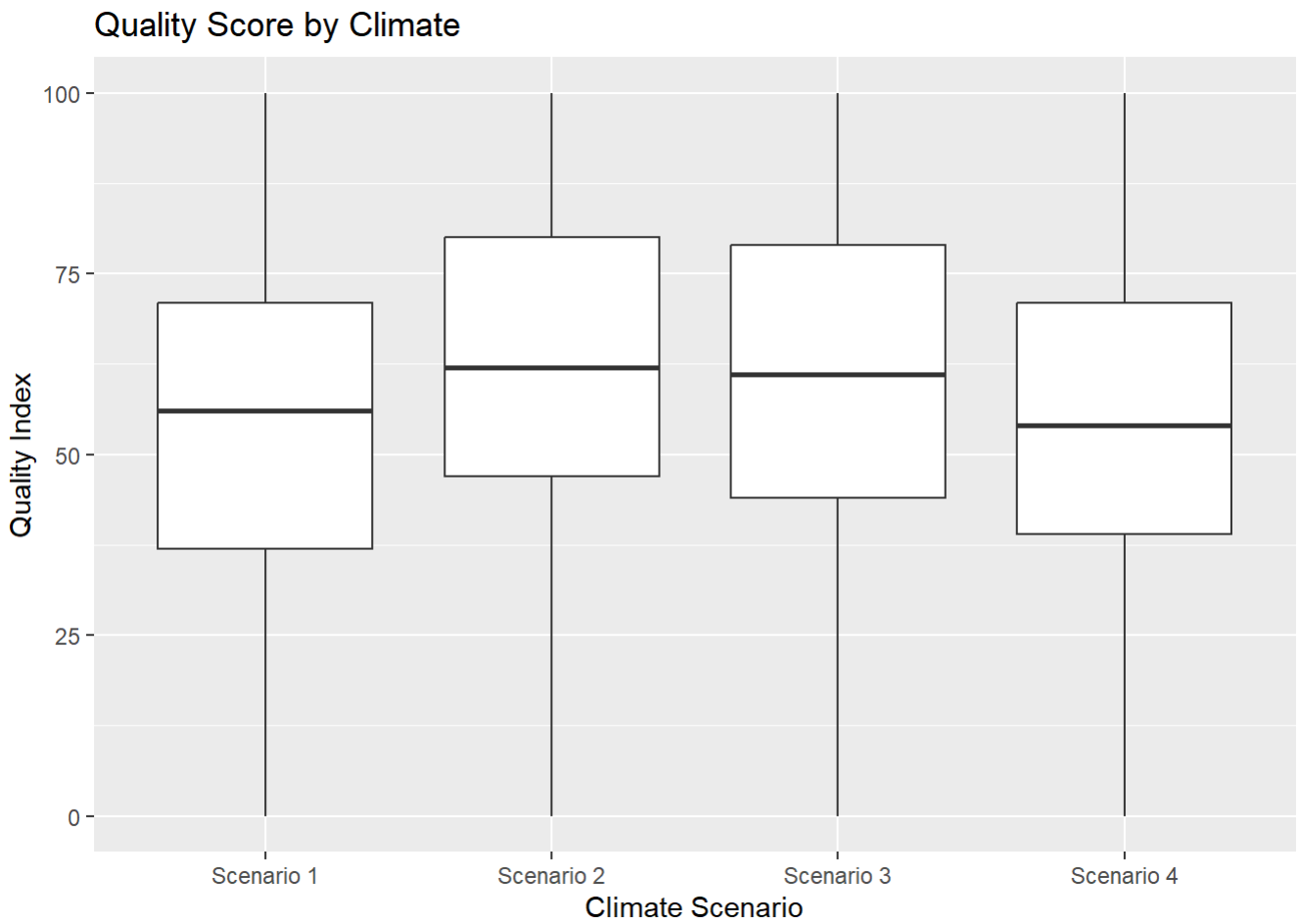
```r
# Missing values
missing_summary <- colSums(is.na(outcome))
missing_summary
```

```
##          ID    location    ecotr_id     tree_id     pear_id     species
##           0           0           0           0           0           0
##     climate quality_idx
##           0           0
```

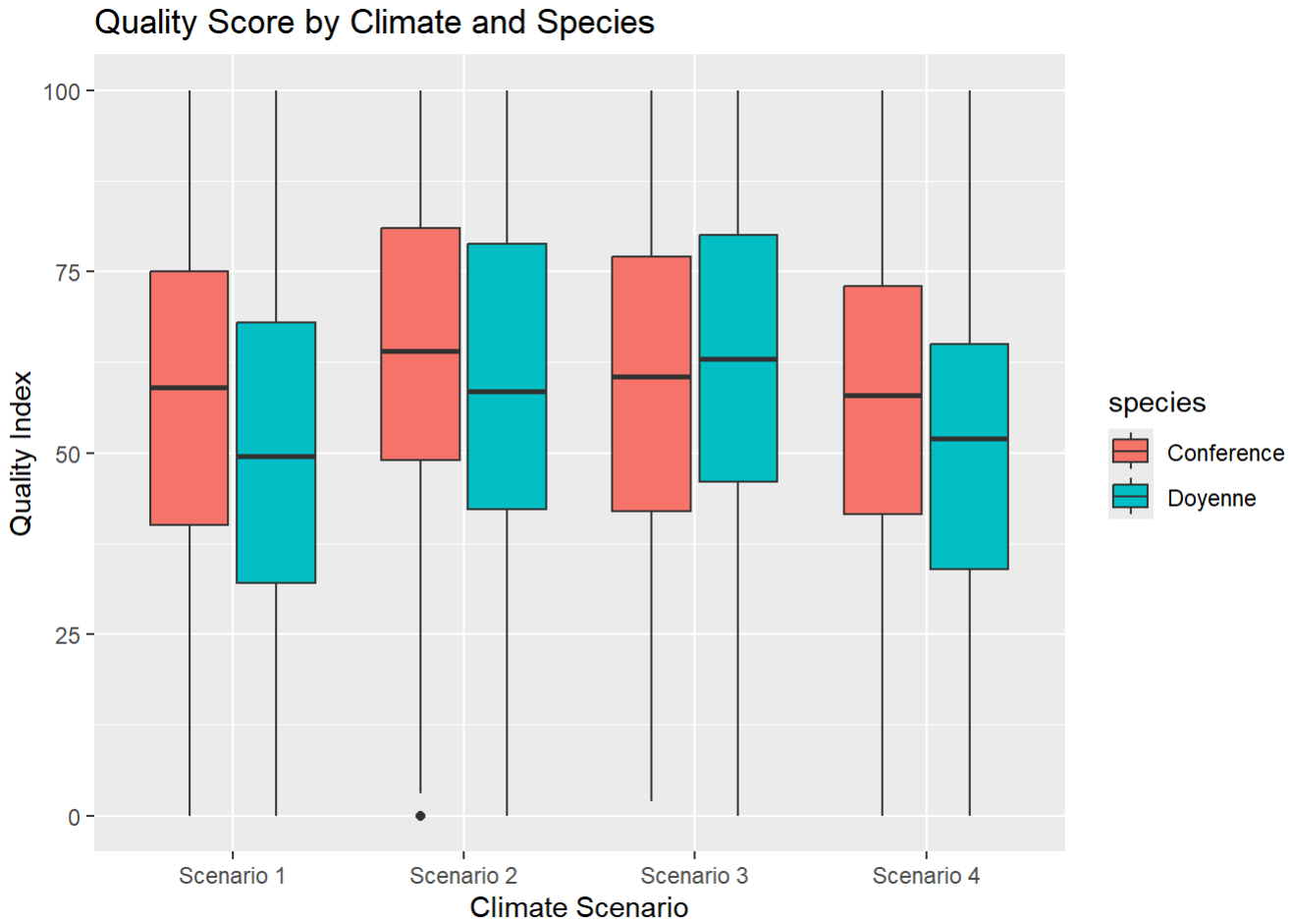
```
# Distribution of quality score
ggplot(outcome, aes(x = quality_idx)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Quality Score",
    x = "Quality Index",
    y = "Count"
  )
```




```
# Boxplot by climate
ggplot(outcome, aes(x = climate, y = quality_idx)) +
  geom_boxplot() +
  labs(
    title = "Quality Score by Climate",
    x = "Climate Scenario",
    y = "Quality Index"
  )
```




```
# Boxplot by climate and species
ggplot(outcome, aes(x = climate, y = quality_idx, fill = species)) +
  geom_boxplot() +
  labs(
    title = "Quality Score by Climate and Species",
    x = "Climate Scenario",
    y = "Quality Index"
  )
```



Inferential model (Should be the primary modedl)

```
# Set reference categories
outcome$climate <- relevel(outcome$climate, ref = "Scenario 1")
outcome$species <- relevel(outcome$species, ref = "Conference")

# Fit model
m_primary <- lmer(
  quality_idx ~ climate * species + (1 | tree_id),
  data = outcome,
  REML = TRUE
)

# Output
summary(m_primary)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: quality_idx ~ climate * species + (1 | tree_id)
##    Data: outcome
##
## REML criterion at convergence: 15494.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.66245 -0.70291  0.01687  0.73495  2.13207
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  tree_id  (Intercept)   3.388   1.841
##  Residual             553.515  23.527
## Number of obs: 1695, groups:  tree_id, 72
##
## Fixed effects:
##                                  Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)                       57.8242     1.8234 82.2031  31.712  < 2e-16
## climateScenario 2                  6.2819     2.3442 57.1283   2.680  0.00961
## climateScenario 3                  1.8815     2.4958 73.6446   0.754  0.45334
## climateScenario 4                 -0.9577     2.5713 82.7680  -0.372  0.71049
## speciesDoyenne                    -7.3918     2.6785 96.8401  -2.760  0.00692
## climateScenario 2:speciesDoyenne   2.6400     3.4409 67.3510   0.767  0.44561
## climateScenario 3:speciesDoyenne   9.6922     3.6352 83.9401   2.666  0.00920
## climateScenario 4:speciesDoyenne   1.7968     3.7803 96.7782   0.475  0.63565
##
## (Intercept)                      ***
## climateScenario 2                **
## climateScenario 3
## climateScenario 4
## speciesDoyenne                   **
## climateScenario 2:speciesDoyenne
## climateScenario 3:speciesDoyenne **
## climateScenario 4:speciesDoyenne
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) clmtS2 clmtS3 clmtS4 spcsDy clS2:D clS3:D
## climatScnr2 -0.778
## climatScnr3 -0.731  0.568
## climatScnr4 -0.709  0.552  0.518
## speciesDynn -0.681  0.530  0.497  0.483
## clmtS2:spcD  0.530 -0.681 -0.387 -0.376 -0.778
## clmtS3:spcD  0.502 -0.390 -0.687 -0.356 -0.737  0.574
## clmtS4:spcD  0.482 -0.375 -0.352 -0.680 -0.709  0.552  0.522
```

```
anova(m_primary, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## climate         18918.0  6306.0     3 71.390 11.3926 3.435e-06 ***
## species          5277.7  5277.7     1 75.361  9.5349  0.002822 **
## climate:species  4761.5  1587.2     3 71.390  2.8674  0.042469 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Predictive model (secondary)

5-Fold CV models

```
# Set references
outcome$climate <- relevel(outcome$climate, ref = "Scenario 1")
outcome$species <- relevel(outcome$species, ref = "Conference")

# Check
table(outcome$climate, outcome$species)
```

```
##
##              Conference Doyenne
##   Scenario 1        189     160
##   Scenario 2        311     262
##   Scenario 3        220     201
##   Scenario 4        191     161
```

```
m_base <- lmer(
  quality_idx ~ climate * species + (1 | tree_id),
  data = outcome,
  REML = TRUE
)

summary(m_base)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: quality_idx ~ climate * species + (1 | tree_id)
##    Data: outcome
##
## REML criterion at convergence: 15494.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.66245 -0.70291  0.01687  0.73495  2.13207
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  tree_id  (Intercept)   3.388   1.841
##  Residual             553.515  23.527
## Number of obs: 1695, groups:  tree_id, 72
##
## Fixed effects:
##                                  Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)                       57.8242     1.8234 82.2031  31.712  < 2e-16
## climateScenario 2                  6.2819     2.3442 57.1283   2.680  0.00961
## climateScenario 3                  1.8815     2.4958 73.6446   0.754  0.45334
## climateScenario 4                 -0.9577     2.5713 82.7680  -0.372  0.71049
## speciesDoyenne                    -7.3918     2.6785 96.8401  -2.760  0.00692
## climateScenario 2:speciesDoyenne   2.6400     3.4409 67.3510   0.767  0.44561
## climateScenario 3:speciesDoyenne   9.6922     3.6352 83.9401   2.666  0.00920
## climateScenario 4:speciesDoyenne   1.7968     3.7803 96.7782   0.475  0.63565
##
## (Intercept)                      ***
## climateScenario 2                **
## climateScenario 3
## climateScenario 4
## speciesDoyenne                   **
## climateScenario 2:speciesDoyenne
## climateScenario 3:speciesDoyenne **
## climateScenario 4:speciesDoyenne
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) clmtS2 clmtS3 clmtS4 spcsDy clS2:D clS3:D
## climatScnr2 -0.778
## climatScnr3 -0.731  0.568
## climatScnr4 -0.709  0.552  0.518
## speciesDynn -0.681  0.530  0.497  0.483
## clmtS2:spcD  0.530 -0.681 -0.387 -0.376 -0.778
## clmtS3:spcD  0.502 -0.390 -0.687 -0.356 -0.737  0.574
## clmtS4:spcD  0.482 -0.375 -0.352 -0.680 -0.709  0.552  0.522
```

```
anova(m_base, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## climate         18918.0  6306.0     3 71.390 11.3926 3.435e-06 ***
## species          5277.7  5277.7     1 75.361  9.5349  0.002822 **
## climate:species  4761.5  1587.2     3 71.390  2.8674  0.042469 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
set.seed(123)

folds <- group_vfold_cv(
  outcome,
  group = tree_id,
  v = 5
)
```

```
candidate_models <- list(
  m1 = quality_idx ~ climate + species + (1 | tree_id),
  m2 = quality_idx ~ climate * species + (1 | tree_id)
)
```

```
fit_and_evaluate <- function(split, formula) {

  train <- analysis(split)
  test  <- assessment(split)

  model <- lmer(
    formula,
    data = train,
    REML = FALSE
  )

  preds <- predict(
    model,
    newdata = test,
    allow.new.levels = TRUE
  )

  tibble(
    rmse = rmse_vec(test$quality_idx, preds),
    mae  = mae_vec(test$quality_idx, preds),
    rsq  = rsq_vec(test$quality_idx, preds)
  )
}
```

```r
cv_results <- map_dfr(names(candidate_models), function(name) {

  formula <- candidate_models[[name]]

  results <- map_dfr(folds$splits, fit_and_evaluate, formula = formula)

  results %>%
    mutate(model = name, .before = 1)
})
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```r
cv_results
```

```
## # A tibble: 10 × 4
##    model  rmse   mae     rsq
##    <chr> <dbl> <dbl>   <dbl>
##  1 m1     24.4  19.8 0.0127
##  2 m1     22.9  18.7 0.0321
##  3 m1     23.5  18.9 0.0188
##  4 m1     22.8  18.9 0.0127
##  5 m1     24.5  20.1 0.0360
##  6 m2     24.5  19.9 0.00871
##  7 m2     22.8  18.6 0.0433
##  8 m2     23.5  19.0 0.0230
##  9 m2     22.8  18.9 0.0125
## 10 m2     24.5  19.9 0.0419
```

```r
cv_summary <- cv_results %>%
  group_by(model) %>%
  summarise(
    mean_rmse = mean(rmse),
    sd_rmse   = sd(rmse),
    mean_mae  = mean(mae),
    mean_rsq  = mean(rsq),
    .groups = "drop"
  ) %>%
  arrange(mean_rmse)

cv_summary
```

```
## # A tibble: 2 × 5
##   model mean_rmse sd_rmse mean_mae mean_rsq
##   <chr>     <dbl>   <dbl>    <dbl>    <dbl>
## 1 m2         23.6   0.852     19.3   0.0259
## 2 m1         23.6   0.815     19.3   0.0224
```

```
best_model_name <- cv_summary %>%
  slice(1) %>%
  pull(model)

best_model_name
```

```
## [1] "m2"
```

```
final_formula <- candidate_models[[best_model_name]]

m_final <- lmer(
  final_formula,
  data = outcome,
  REML = TRUE
)

summary(m_final)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: final_formula
##    Data: outcome
##
## REML criterion at convergence: 15494.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.66245 -0.70291  0.01687  0.73495  2.13207
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
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## climateScenario 4                 -0.9577     2.5713 82.7680  -0.372  0.71049
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##
## (Intercept)                      ***
## climateScenario 2                **
## climateScenario 3
## climateScenario 4
## speciesDoyenne                   **
## climateScenario 2:speciesDoyenne
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## clmtS3:spcD  0.502 -0.390 -0.687 -0.356 -0.737  0.574
## clmtS4:spcD  0.482 -0.375 -0.352 -0.680 -0.709  0.552  0.522
```